Getting Started with Python

For `COM6018` students

Varvara Papazoglou

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About me

- Varvara Papazoglou (email address is v.papazoglou@sheffield.ac.uk)
- University Teaching Associate
- Research interests
  - Natural Language Processing and Music Information Retrieval
    - Song lyrics, music playlists, metadata, user studies
    - Music recommendation, topic classification/clustering, human evaluation
  - Artificial Intelligence in Higher Education
    - AI literacy, educational software
- You can find out more on my webpage.

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Purpose of this session

`COM6018 Data Science with Python` will be taught in the Bring Your Own Device (BYOD) format.

You will be using your own laptop for the sessions.

This session is designed to set you up with the tools you will need.

- A code editor (Visual Studio Code)
- Python (with `uv` to manage versions and environments)
- Git and GitHub (for version control and code sharing)

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Session outline

1. Installing Visual Studio Code
2. Installing Python with `uv`
3. Developing with Git
4. Using GitHub
5. Setting up your first Python project
6. Building a Cow-themed ChatBot 🐮

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Session materials

All the materials for this session are online at

https://github.com/UOS-COM6018-2627/COM6018

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What is Git?

- Git is an open-source, free `version control system`
- It allows you to track changes to files in a database called a `repository`
- It allows you to revert to previous versions
- It allows you to work on different versions of files at the same time (branches)

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What is Git?

- It is a `distributed` version control system which means local copies of the repository can be kept on different computers.
- You can work offline and then `push` your changes to a remote repository when you are ready or you can `pull` changes that others have made.

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What is GitHub?

- GitHub is a `hosting service` for Git repositories that can make repositories available to others over the internet.
- It is a commercial service but free for open-source projects
- It provides a web interface for viewing and editing repositories
- It also provides many other features such as issue tracking, wikis, project management, etc.

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A very brief introduction to Git

- Installing Git
- Making a new repository
- Saving file changes to a repository
- Reverting to an earlier version
- Saving your repository on GitHub

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Installing Git

- Git is freely available for Windows, Mac and Linux
- To install visit https://git-scm.com/downloads
- Latest version at time of writing is 2.55.0
- (Windows Git comes with Git Bash, a Unix-like terminal that can be used to run Git commands.)

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Making a new Git repository

A Git repository can track files in a directory and its subdirectories.

To make a repository for your project, navigate to the directory and type:

```
cd my_project
git init
```

This will make an empty repository. You now need to add (`commit`) files to the repository.

```
# Use `add .` to add all the files in the directory tree
git add .
# The -m flag allows us to provide a `commit message`
git commit -m "Initial commit"
```

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Updating a Git repository

After we've made changes to some files, or added/deleted files, we can save the new state to the repository.

```
# Use the -u flag to add all modified files
git add -u

# Use a meaningful commit message
git commit -m "Added the results analysis"
```

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View the history of a Git repository

We can view the history of a repository using `git log`

```
git log
```

This will show something like the following,

```
commit 56e41739ec18f8b88b9dc7321147068d40244096 (HEAD)
Author: Varvara Papazoglou <v.papazoglou@sheffield.ac.uk>
Date:   Sat Sep 30 18:37:10 2023 +0100

    Added the results analysis

commit e995371ea26d77006cbb2e33268ef2346a9e7c1a
Author: Varvara Papazoglou <v.papazoglou@sheffield.ac.uk>
Date:   Sat Sep 30 18:27:33 2023 +0100

    Initial commit
```

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Reverting to a previous version

We can revert to a previous state of the repository.

We need to know the `commit hash` of the version we want to revert to. This is the long string of characters on the first line of the `git log` output. e.g. 'e995371...'

The we can temporarily revert to that version using `git checkout`

```
git checkout e995371
```

Or permanently revert to that version using `git reset`

```
git reset --hard e995371
```

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Cloning a repository

We can make a copy of an existing repository using `git clone`

```
git clone <url>
```

This will make a copy of the repository in a new directory with the same name as the repository.

We can easily clone repositories from GitHub. For example, to clone the COM6018 repository:

```
git clone https://github.com/UOS-COM6018-2627/COM6018.git
```

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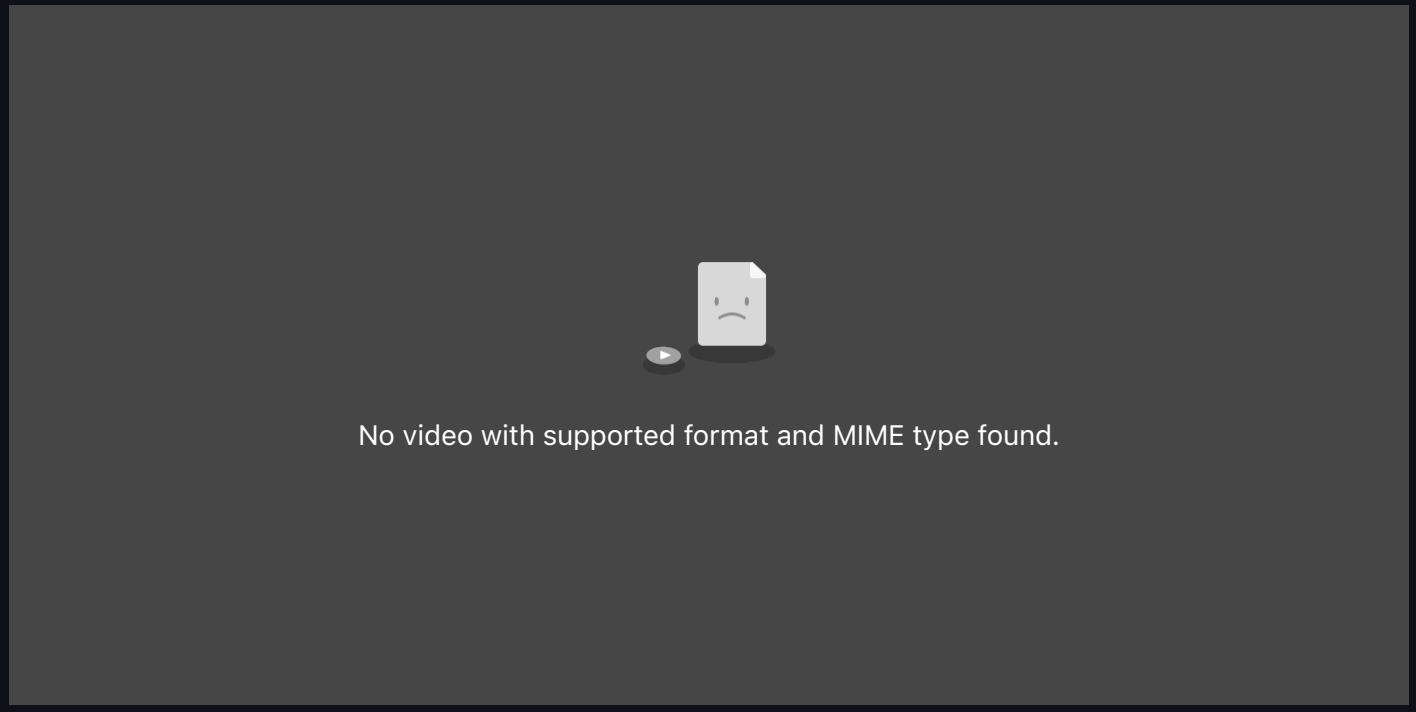
Saving your own repository on GitHub

To save your own repository on GitHub you need to,

1. Make yourself an account on GitHub, (visit https://github.com)
2. create a new empty repository on GitHub
3. `push` your local repository to GitHub.

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Demo



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